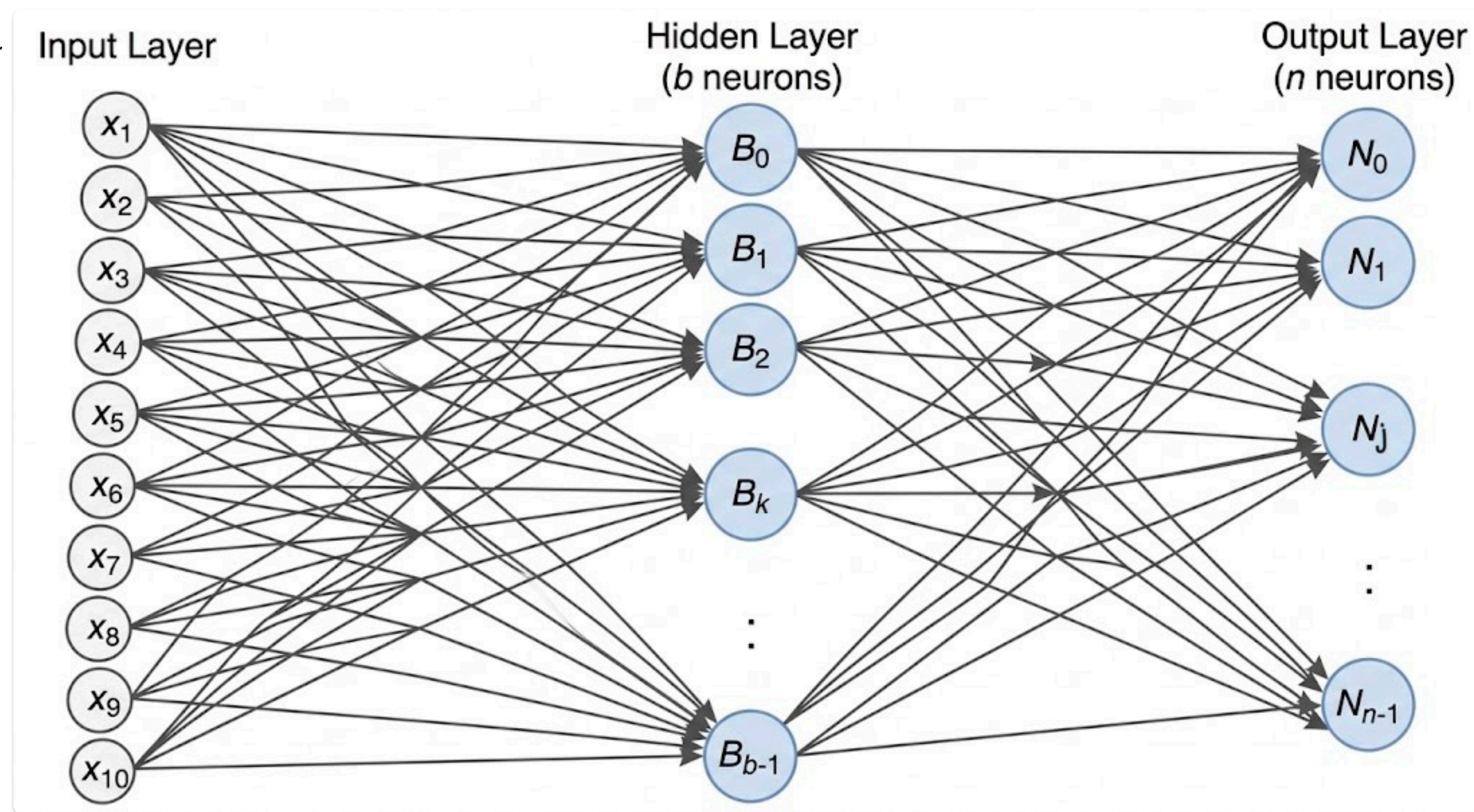


1. Loss Function for Multiclass Classification:

A.



B. forward pass — error function:

$$\text{error function} = e = \text{Mean Square Error (MSE)} = \frac{1}{m} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} (y_{ij} - \hat{y}_{ij})^2$$

y_{ij} = desire output y (ground truth) from i -th pair training example which corresponding to j -th output perceptron N .
 $\hat{y}_{ij}$ = predicted output $\hat{y}$ from j -th output perceptron N which corresponding to i -th pair training example.
 m = number of pair training examples
 n = number of output perceptrons N in the neural network.

C. backward pass — Overall Error of j -th output (Overall Error $_j$):

⇒ Overall Error of j -th output corresponding i -th pair training example (Overall Error $_{ij}$)

⇒ e partial derivate $\hat{y}_{ij}$

$$\Rightarrow e'(\hat{y}_{ij}) = \frac{\partial e}{\partial \hat{y}_{ij}} = -\frac{2}{m} (y_{ij} - \hat{y}_{ij})$$

$$\Rightarrow \text{Overall Error}_{ij} = -\frac{2}{m} (y_{ij} - \hat{y}_{ij})$$

$$\Rightarrow \text{Overall Error}_j = \sum_{i=0}^{m-1} \text{Overall Error}_{ij}$$

D. backward pass — Error of j -th output perceptron N (Error N_j):

⇒ Error of j -th output perceptron N corresponding i -th pair training example (Error N_{ij})

⇒ e partial derivate $z_{N_{ij}}$ ($z_{N_{ij}}$ = weighted sum function of j -th output perceptron N corresponding i -th pair training example)

$$\Rightarrow \text{Error } N_{ij} = \frac{\partial e}{\partial z_{N_{ij}}} = \frac{\partial e}{\partial f_{N_{ij}}} \cdot \frac{\partial f_{N_{ij}}}{\partial z_{N_{ij}}} \quad (f_{N_{ij}} = \text{activation function of } j\text{-th output perceptron } N \text{ corresponding } i\text{-th pair training example} = \hat{y}_{ij})$$

$$\Rightarrow \text{Error } N_{ij} = \frac{\partial e}{\partial z_{N_{ij}}} = \frac{\partial e}{\partial \hat{y}_{ij}} \cdot \frac{\partial \hat{y}_{ij}}{\partial z_{N_{ij}}} = (\text{Overall Error}_{ij}) \cdot f'_{N_{ij}}(z_{N_{ij}})$$

$$\Rightarrow \text{Error } N_j = \sum_{i=0}^{m-1} \text{Error } N_{ij}$$

E. backward pass — Error of k -th hidden perceptron B (Error B_k):

⇒ Error of k -th hidden perceptron B corresponding i -th pair training example (Error B_{ik})

⇒ e partial derivate $z_{B_{ik}}$ ($z_{B_{ik}}$ = weighted sum function of k -th hidden perceptron B corresponding i -th pair training example)

$$\Rightarrow \text{Error } B_{ik} = \frac{\partial e}{\partial z_{B_{ik}}} = \left(\frac{\partial e}{\partial f_{N_{i0}}} \cdot \frac{\partial f_{N_{i0}}}{\partial z_{N_{i0}}} \cdot \frac{\partial z_{N_{i0}}}{\partial f_{B_{ik}}} + \frac{\partial e}{\partial f_{N_{i1}}} \cdot \frac{\partial f_{N_{i1}}}{\partial z_{N_{i1}}} \cdot \frac{\partial z_{N_{i1}}}{\partial f_{B_{ik}}} + \frac{\partial e}{\partial f_{N_{i2}}} \cdot \frac{\partial f_{N_{i2}}}{\partial z_{N_{i2}}} \cdot \frac{\partial z_{N_{i2}}}{\partial f_{B_{ik}}} \dots + \frac{\partial e}{\partial f_{N_{i(n-1)}}} \cdot \frac{\partial f_{N_{i(n-1)}}}{\partial z_{N_{i(n-1)}}} \cdot \frac{\partial z_{N_{i(n-1)}}}{\partial f_{B_{ik}}} \right) \cdot \frac{\partial f_{B_{ik}}}{\partial z_{B_{ik}}} \quad (f_{B_{ik}} = \text{activation function of } k\text{-th hidden perceptron } B \text{ corresponding } i\text{-th pair training example})$$

$$\Rightarrow \text{Error } B_{ik} = (\text{Error } N_{i0} \cdot w_{N_{i0}} + \text{Error } N_{i1} \cdot w_{N_{i1}} + \text{Error } N_{i2} \cdot w_{N_{i2}} \dots + \text{Error } N_{i(n-1)} \cdot w_{N_{i(n-1)}}) \cdot f'_{B_{ik}}(z_{B_{ik}})$$

$$\Rightarrow \text{Error } B_{ik} = \left(\sum_{j=0}^{n-1} \text{Error } N_{ij} \cdot w_{N_{ij}} \right) \cdot f'_{B_{ik}}(z_{B_{ik}}) \quad (w_{N_{ij}} = \text{weight of } j\text{-th output perceptron } N \text{ corresponding } i\text{-th pair training example})$$

$$\Rightarrow \text{Error } B_k = \sum_{i=0}^{m-1} \text{Error } B_{ik}$$

F. weight adjustment — all weights in the neural network:

⇒ Gradient Descent Method: $\vec{W}' = \vec{W} + \eta \cdot (-\nabla e)$

⇒ Gradient Descent Method: $\vec{W}' = \vec{W} + \eta \cdot (-\text{"Error"} \times \text{"of the weight belonged to } x\text{-th perceptron } Y" \cdot \text{"Input of the weight belonged to } x\text{-th perceptron } Y"})$